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CT60A5401 GAME DEVELOPMENT PROJECT

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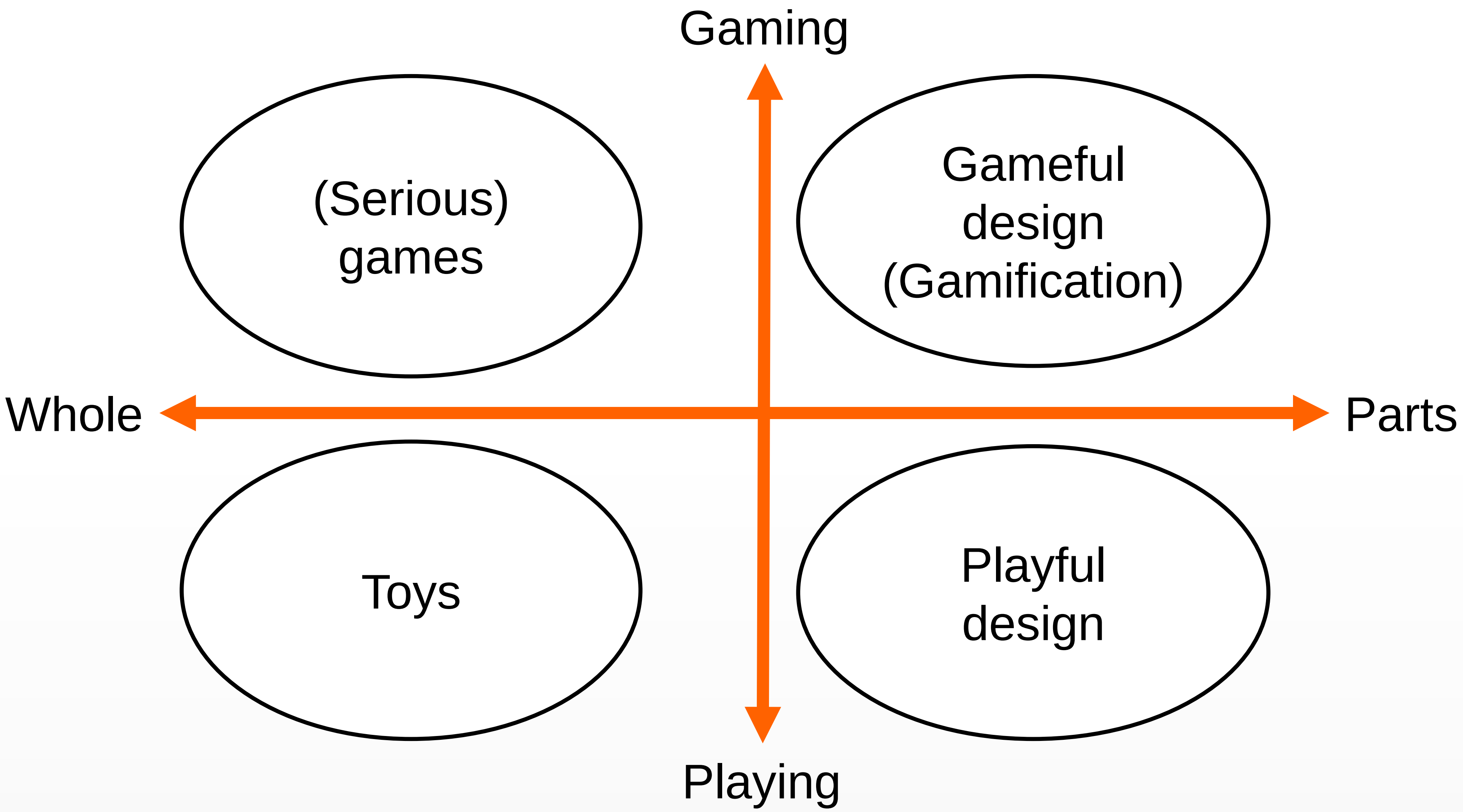
Gamification, Serious Games and Games For Health

- A Crash course into the “serious” side of the game industry

Gamification

- Fundamentally, a catch-all term describing the act of bringing game-like elements into a system, which traditionally may not apply such things.
- In more technical terms: *"Gamification is a form of service packaging where a core service is enhanced by a rules-based service system that provides feedback and interaction mechanisms to the user with an aim to facilitate and support the users‟ overall value creation"* (Huotari and Hamari 2013)


Gaming
(Serious) games
Gameful design (Gamification)
Whole
Parts
Toys
Playful design
Playing

Gamification

- **Gameful design (Gamification)** implies that the software product is designed with the game-like components forming a part of the system design, but the product itself has a functional non-game purpose and elements, which are not game-like.
- **Serious Games (and other games)** are products, which are fully built from the game components and game-like elements. If the product has a “real” purpose, it can be classified as a Serious Game, otherwise normal – entertainment- game.
- **Playful design** implies a system, which has playful elements in its design, but also has components which are not playful and the system has a non-play purpose.
- **Toys** are products which are fully designed for enabling use in play, and do not have an intended non-play purpose beyond entertainment.

The “traditional” software products are not visible in this model, since they are not designed for gaming or playing, and they only have the functional, non-gamified purpose.

Game Development and Software Development

Obviously, game development and software development share some common features.

- Design work
- Development, programming tasks
- Tools
- Testing
- Objectives:
  - In time
  - In budget
  - With the intended features
  - With acceptable quality

On Game Development

Obviously there also are some differences.

- More emphasis on customers, market
- Design with artistic vision on how things happen.
- Tools not necessarily the same as in software development.
- Testing with users.
- Sound design.

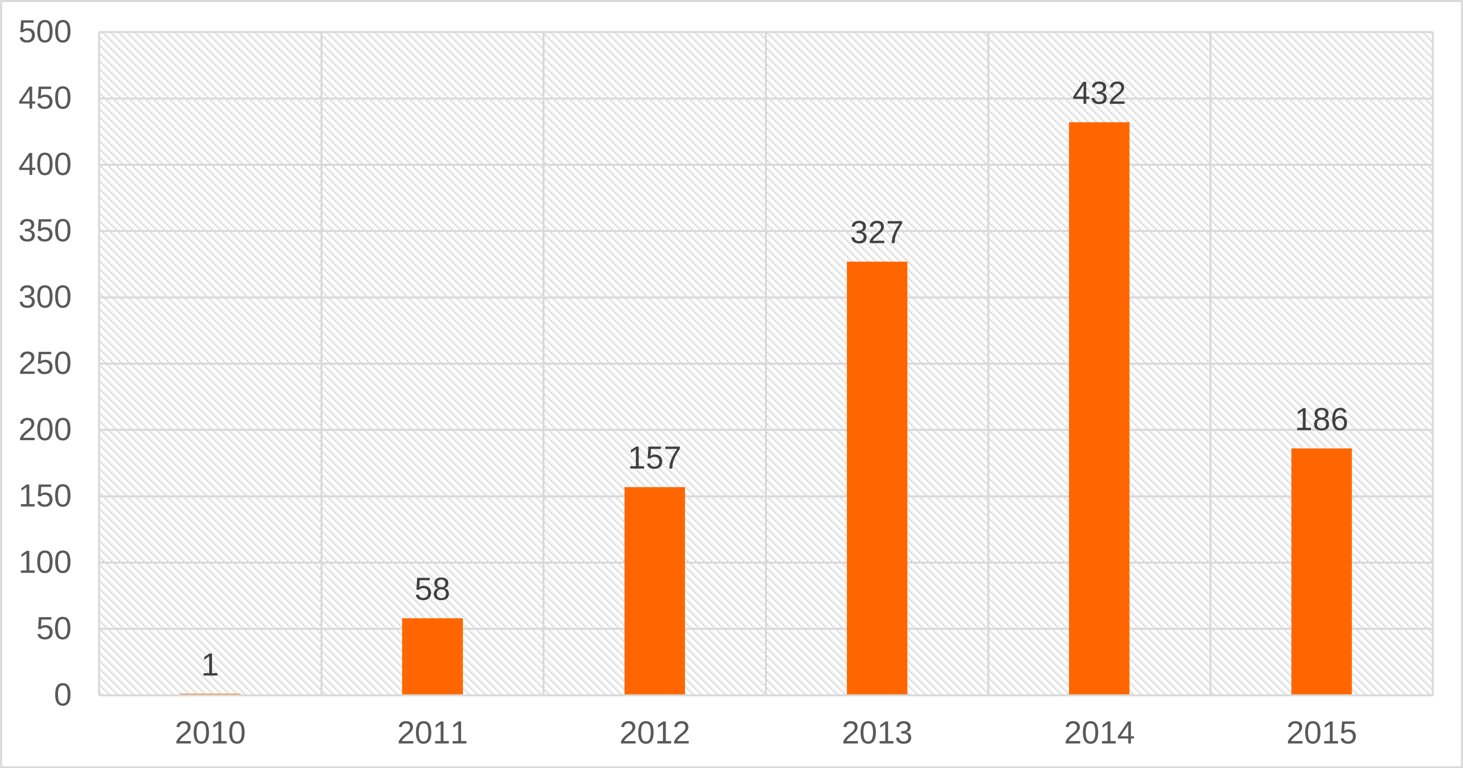
On Game Development

- Development has certain aspects which are unusual:
  - Real time client-to-client multiplayer systems
  - Artificial intelligence
  - 3D mathematics
- Objectives:
  - In time, in budget, acceptable quality, all features, and.
    - Is the game fun?
    - Is the product engaging?

On Gamification design

- Goals
  - User retention rate
  - Promote motivation
  - Longer sessions
    - Indepth interest
    - More frequent visits
- Do the gamified elements add something valuable to the user experience?

Gamification (popularity as an amount of published scientific works, data Sept 2015)



Gamification (results from a survey into the study topics)

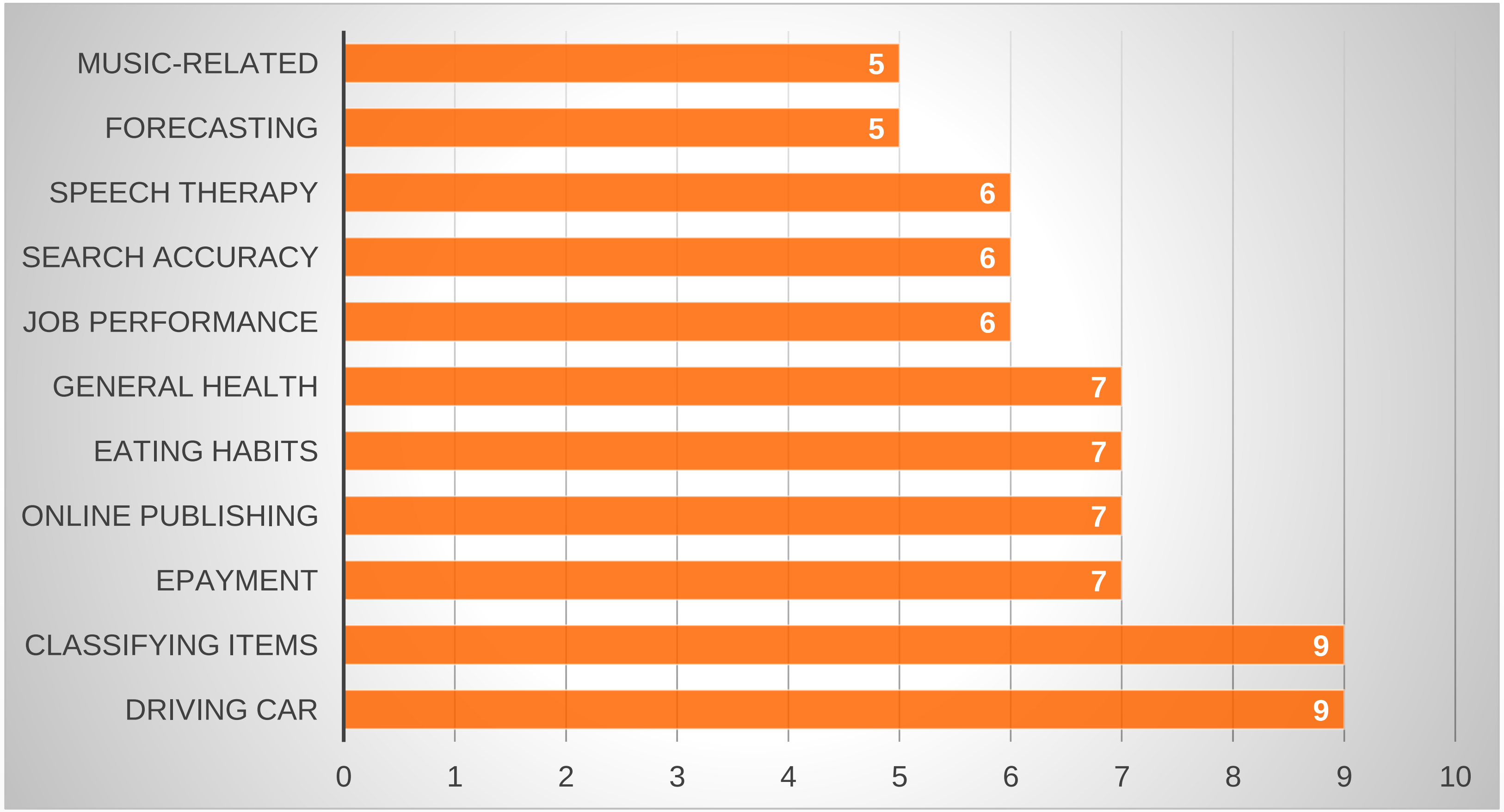
- Some common areas of research in gamification:
  - General theory and elements (21.6%)
  - Different Proof-of-Concept-studies (26.5%)
  - Serious Games (2.6%)
  - Games For Health (9.4%)
  - Crowdsourcing applications (8.9%)
  - Business studies (4.3 %)
  - Technology (Hardware) studies (4.9 %)
  - MOOC- or eLearning-related (17.7%)

Rest “Others” like literature reviews, abstracts, posters, keynote summaries etc.

Domains of Gamification Proof-of-Concepts

POC – Computer Science teaching tool	29 (out of 308)	9.4 % (out of 308)	The presented application focuses on teaching computer science discipline; for example learning programming, or learning test case design.
POC – Ecologically Friendly Lifestyle Teaching tool	25 (out of 308)	8.1 % (out of 308)	The presented application focuses on teaching ecologically friendlier lifestyle; for example conserving electricity, anti-littering behavior or otherwise preserving the environment.
POC – Motivation Improvement tool	25 (out of 308)	8.1 % (out of 308)	The presented application focuses on improving the motivation for conducting some task; for example classifying images or completing given task lists.
POC – Software Development tool	23 (out of 308)	7.4 % (out of 308)	The presented application focuses on improving the motivation to do software development; for example identifying requirements or participating on design work.
POC – Social Behavior	21 (out of 308)	6.8 % (out of 308)	The presented application focuses on improving behavior, or participation of users.
POC – Non-CS STEM-topic Teaching tool	14 (out of 308)	4.5 % (out of 308)	The presented application focuses on teaching science, technology, engineering or medicine, but not computer science or closely related field.
POC – eGovernment	11 (out of 308)	3.6 % (out of 308)	The presented application focuses on empowering citizens, to participate on the government activities or report problems.
POC –Business Management tool	11 (out of 308)	3.6 % (out of 308)	The presented application focuses on business management; for example participation to meetings, or logistics management.
POC – Physiotherapy Self-Training tools	11 (out of 308)	3.6 % (out of 308)	The presented application focuses on helping patients to do independent physiotherapy training.
POC – History and museums	10 (out of 308)	3.2 % (out of 308)	The presented application focuses on application on learning history, or participating in activities built into museums.

Further Domains of Gamification Proof-of-Concepts



Crowdsourcing

- González-Ladrón-de-Gueva dictates that crowdsourcing is a type of participative online activity, where a group of individuals undertake a tasks of varying difficulty for mutual benefit.
- Basically crowdsourcing is an act of using volunteering human resources to achieve activities such as image labeling,
  - which would be difficult for a computer to handle and
  - laborious for a small group of people to do.
- In gamified context, the system might for example keep a score on the amount and accuracy of the labels each user has given,
- and offer leaderboards or possibility to use some system feature after attaining certain score or status.

Games For Health

- Games for Health are a subset of serious games, where the concept of the game is to improve the health of the user, either
  - via exercise-inducing games or
  - via promoting healthcare-promoting activities.
- The scale of gamification may change from
  - offering differently themed backgrounds or trophies for achieving certain milestones,
  - to a full-fledged game which enables person to train while playing.

MOOC (Massive Open Online Course)

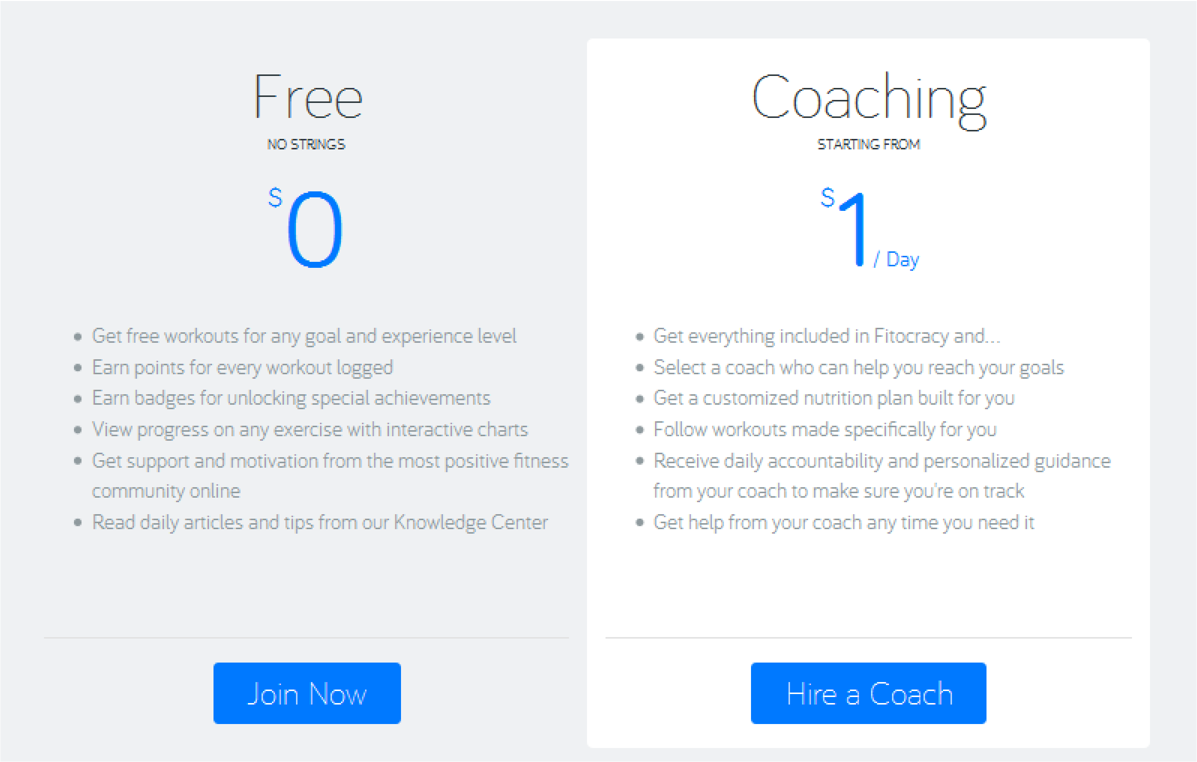
- MOOCs are Internet courses, which are offered to every participant, regardless of their institutional association.
- In MOOC systems, the course contents and participation is open to everyone, but for example official endorsements or graduation diplomas require a separate payment and registration.
- The MOOCs may also involve gamified design in the courseware system to enhance
  - student motivation,
  - retention rate
  - enhance the amount time spent with the course assignments and other self-study material.
- For open, commercial examples, for example CodeCademy offers open, gamified courses on learning introductory-level programming and other computer science topics.

Examples of Gamification

Musical stairs (examples in YouTube)

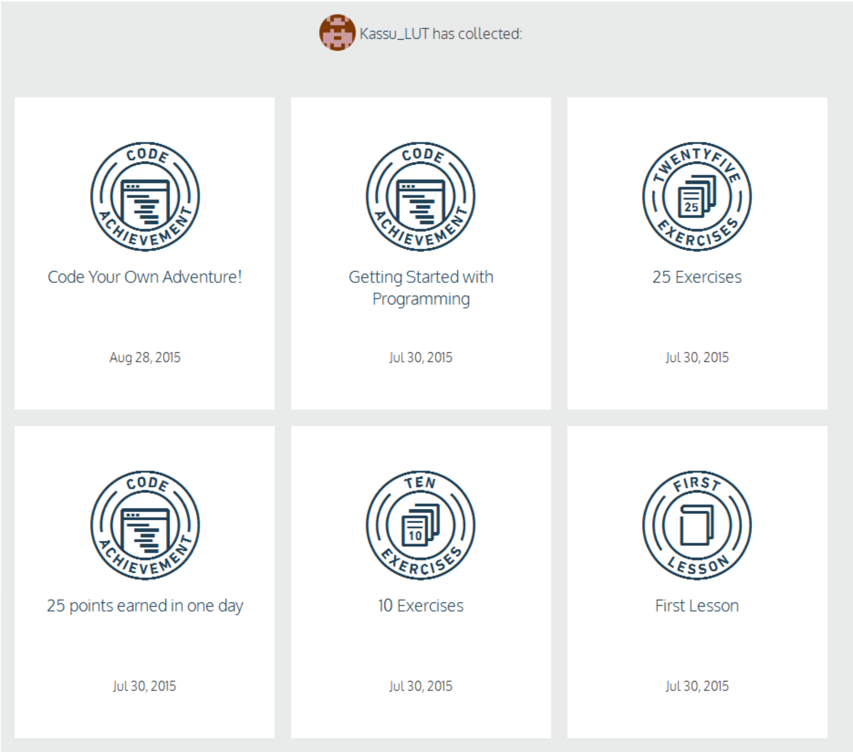
Talking trashcans

Fitocracy



Free
NO STRINGS
$0
Get free workouts for any goal and experience level
Earn points for every workout logged
Earn badges for unlocking special achievements
View progress on any exercise with interactive charts
Get support and motivation from the most positive fitness community online
Read daily articles and tips from our Knowledge Center
Join Now
Coaching
STARTING FROM
$1 / Day
Get everything included in Fitocracy and...
Select a coach who can help you reach your goals
Get a customized nutrition plan built for you
Follow workouts made specifically for you
Receive daily accountability and personalized guidance from your coach to make sure you're on track
Get help from your coach any time you need it
Hire a Coach

Codecademy, Duolingo



Zombies, Run!


Get fit. Escape zombies.
Become a hero!
Collect supplies as you move.
Build and defend your base!

A Couple of Design Pointers (not only for gamification)

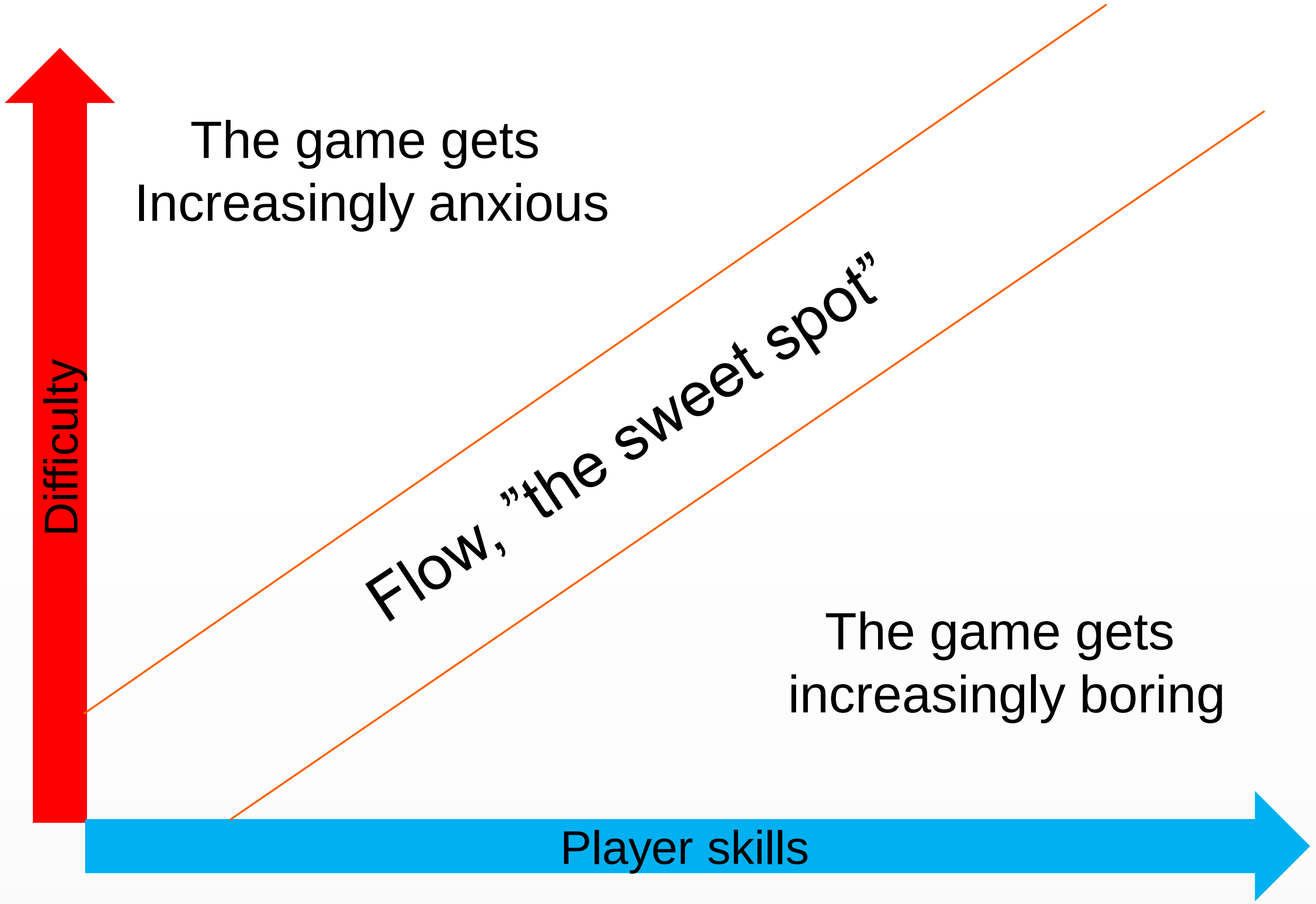
Mostly collected from Werbach & Hunter 2013 and Kapp 2012

Common gamification elements (Werbach&Hunter)

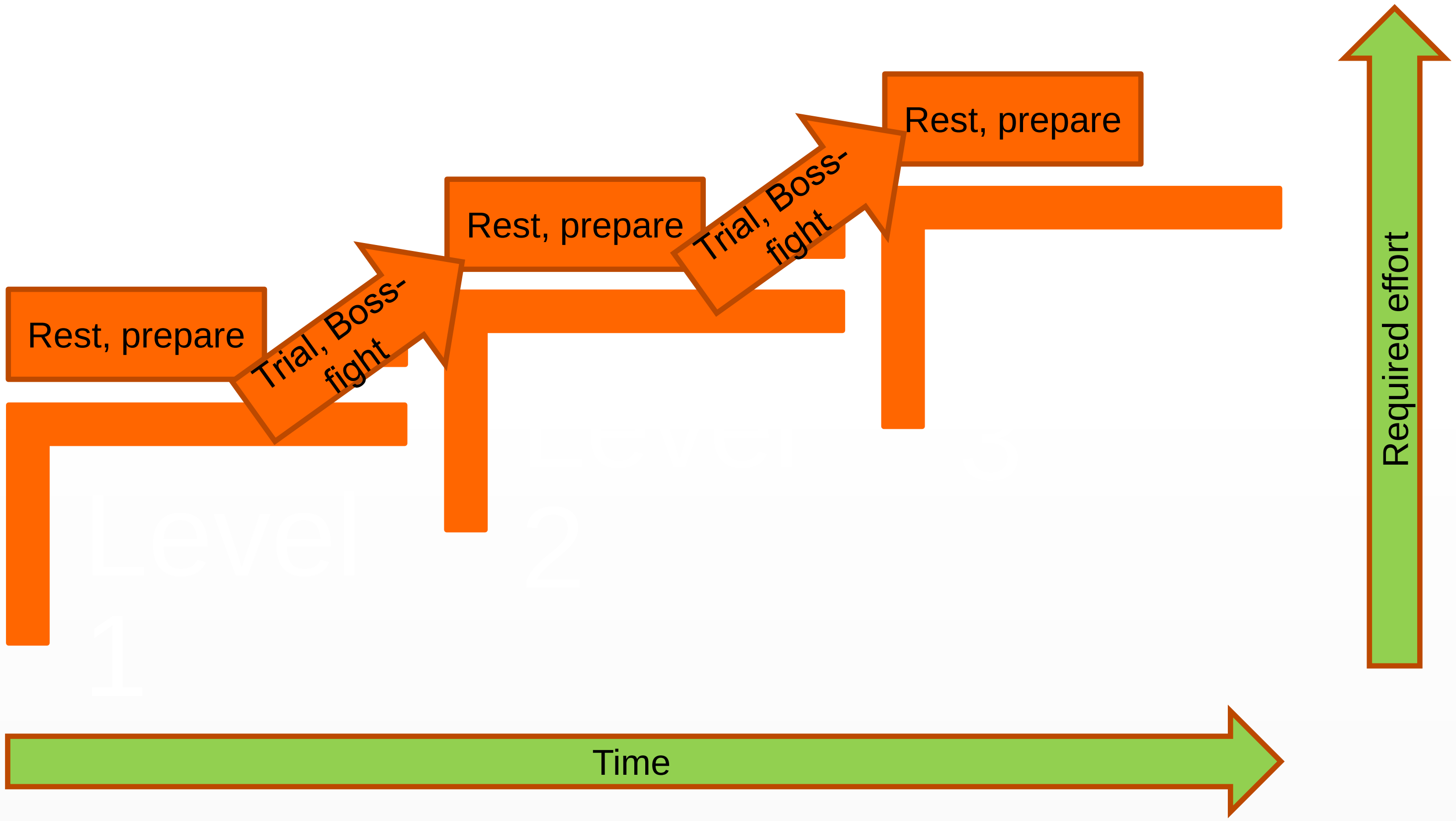
1. Achievements
2. Avatars
3. Badges
4. Boss Fights
5. Collections
6. Combat
7. Content unlocking
8. Gifting
9. Leaderboards
10. Levels
11. Points
12. Quests
13. Social graphs
14. Teams
15. Virtual Goods

Activity Cycle (from Werbach & Hunter)

”Flow” between boredom and anxiety



Progression Stairs



Problematic design

- **Pointsification**: Meaningless addition of points to an existing system.
- **Spotlight stealer**: The game is fun, but no-one is using the actual system as intended.
- **Slave-driver**: Performance measurement by gamification, “Exploitationware”
- **Evil Gamification**: Way to spy or micromanage or compete people (customers, users, employees) against each other in a negative way.
  - Negative aspects for ”losing” the main point
  - Closing service, losing abilities, increasing costs for losing etc.
- **Content controller**: Force users to “enjoy” certain content before allowing them to do what interests them.

What Gamification is not

- **Just badges and achievements**: Elements of games bring new ways to engage users, and allow for example visualization and problem solving as mechanics
- **Trivialization of learning**: The gamification is not used to dilute the learning experience, but to enhance motivation, engagement and retention of students. Real learning is hard, gamification helps with this problem.
- **New concept**: Militaries have had war simulations since the Napoleonic Era. War Games have existed since the 7^{th} century.
- **Easy to implement**: Meaningful gamification takes effort and testing to make it work with the intended users and intended effect.

Well-designed feedback

- Tactile
- Inviting
- Repeatable
- Coherent
- Continuous
- Emergent
- Balanced
- Fresh

So, off to work then!

- Any questions?

- This was the last lecture video, so next steps are to start working on your own game design (project 1), game development (project 2) and when both are done, complete the course exam at the Moodle Pages.

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